

Bottleneck Distance and Stability

Lecture 13 - CMSE 890

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Goals

- Bottleneck distance and stability
- Wasserstein distance

Section 1

Review of setup

Simplicial Complex Filtration from a Function

Definition

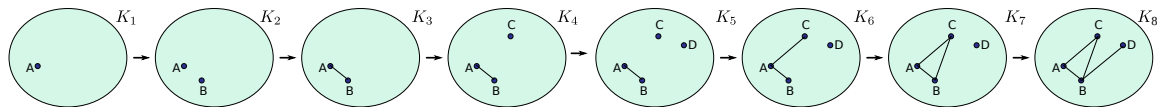
Fix a function $f : K \rightarrow \mathbb{R}$ with $(\sigma \leq \tau \Rightarrow f(\sigma) \leq f(\tau))$

Fix values $a_1 \leq a_2 \leq \dots \leq a_n$

Let $K_i = f^{-1}(-\infty, a_i]$

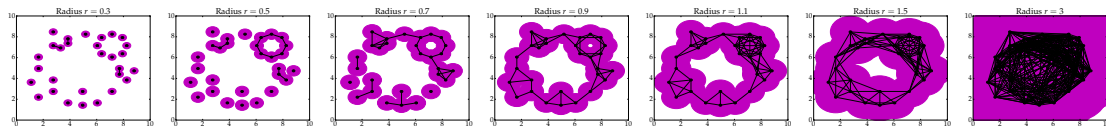
A filtration $\mathcal{F} = \mathcal{F}(K)$ induced by f is a nested sequence of subcomplexes

$$\mathcal{F} : \emptyset = K_0 \subseteq K_1 \subseteq K_2 \subseteq \dots \subseteq K_n = K.$$

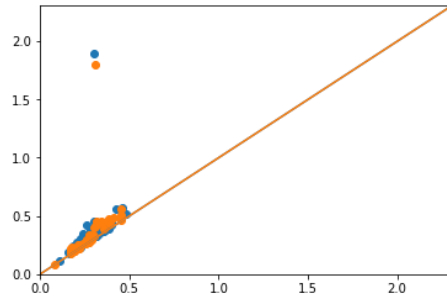
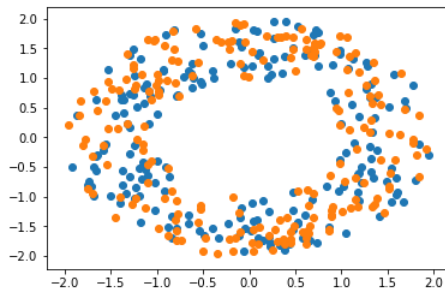


Rips filtration

- Given $P \subseteq \mathbb{R}^d$
- Fix $0 \leq r_1 \leq r_2 \leq \dots \leq r_n$
- Set $K_i = VR(P, r_i)$
- Compute persistence diagram using these radius parameters



Similar persistence diagrams



Section 2

Bottleneck distance

Distance measures

Definition

A distance on a set M is a function

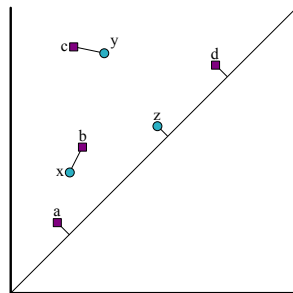
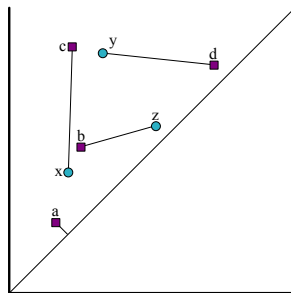
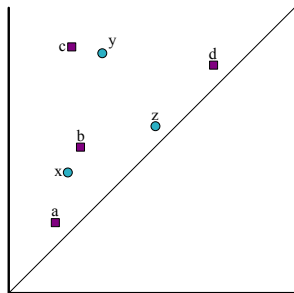
$d : M \times M \rightarrow \mathbb{R}_{\geq 0}$ such that

- $d(x, y) \geq 0$ and
 $d(x, y) = 0$ iff $x = y$
- $d(x, y) = d(y, x)$
- $d(x, z) \leq d(x, y) + d(y, z)$

Ex. $\|(x_1, x_2) - (y_1, y_2)\|_2 =$

Ex. $\|(x_1, x_2) - (y_1, y_2)\|_\infty =$

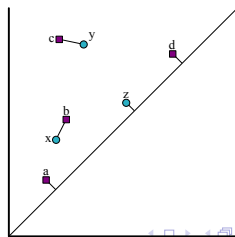
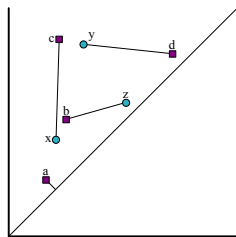
Finding matchings



Definition of bottleneck, Version 1

Given diagrams X and Y , the distance between them is

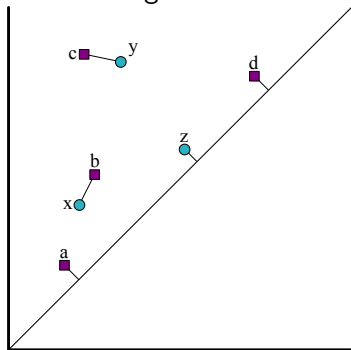
$$d_B(X, Y) = \inf_{\varphi: X \rightarrow Y} \sup_{x \in X} \|x - \varphi(x)\|_\infty$$



Definition of bottleneck, Version 2

Matching

A partial matching between X and Y is a bijection on a subset of the off-diagonal points of the two diagrams $M : X' \rightarrow Y'$ for $X' \subseteq X$ and $Y' \subseteq Y$.

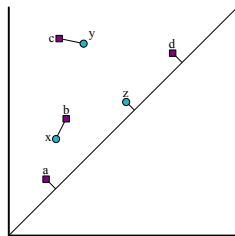


Definition of bottleneck, Version 2

Cost of a matching

The cost of this partial matching is given by

$$c(M) = \sup \left(\{ \|x - M(x)\|_\infty \mid x \in X' \} \cup \left\{ \frac{1}{2} |x_1 - x_2| \mid (x_1, x_2) \in X \setminus X' \cup Y \setminus Y' \right\} \right)$$



Bottleneck distance

Combinatorial definition

The bottleneck distance between X and Y is

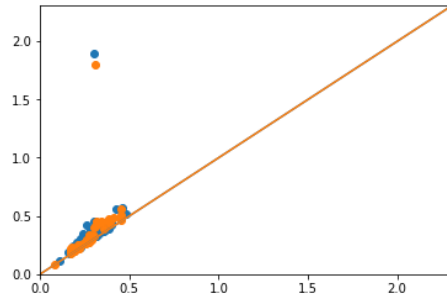
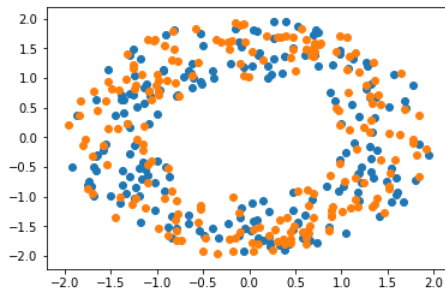
$$d_B(X, Y) = \inf_M c(M)$$

where the infimum runs over all possible partial matchings.

Section 3

Bottleneck stability

Similar persistence diagrams



Stability - Point cloud version

Theorem (Point cloud version)

Let $\mathbb{X}, \mathbb{Y} \subset \mathbb{R}^d$ be a finite point clouds. Let d_H be Hausdorff distance.

Let $D(\mathcal{C}(\mathbb{X}))$ be the persistence diagram of the filtration defined by the Čech complex on $\mathbb{X}$.

Then

$$d_B(\text{Dgm}_p(\mathcal{C}(\mathbb{X})), \text{Dgm}_p(\mathcal{C}(\mathbb{Y}))) \leq d_H(\mathbb{X}, \mathbb{Y}).$$

Stability - Function version

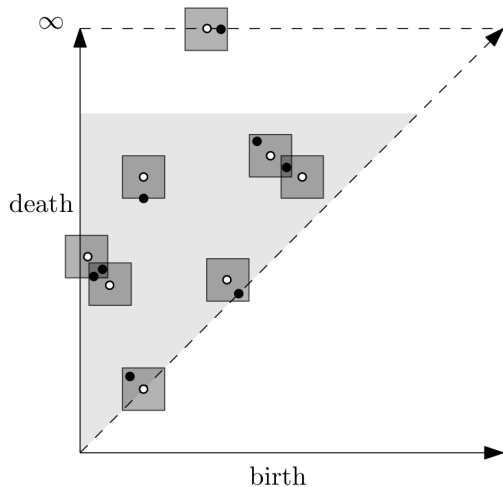
Theorem (Stability for simplicial functions)

Let $f, g : K \rightarrow \mathbb{R}$ be two simplex-wise monotone functions giving rise to two simplicial filtrations $\mathcal{F}_f$ and $\mathcal{F}_g$.

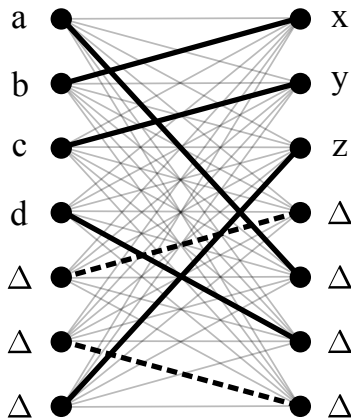
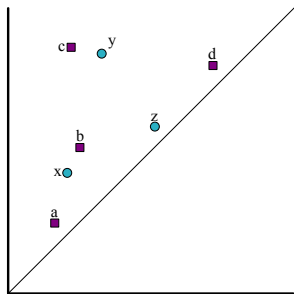
Then, for every $p \geq 0$,

$$d_B(\text{Dgm}_p(\mathcal{F}_f), \text{Dgm}_p(\mathcal{F}_g)) \leq \|f - g\|_\infty.$$

Nearby diagrams



Computing Bottleneck distance



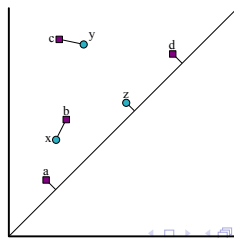
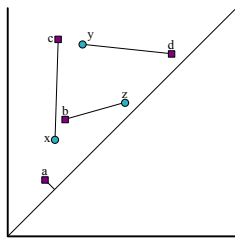
Section 4

Wasserstein Distance and Stability

Wasserstein Distance

Given diagrams X and Y , the p -Wasserstein distance between them is

$$d_{W,p}(X, Y) = \inf_{\varphi: X \rightarrow Y} \left(\sum_{x \in X} \|x - \varphi(x)\|_p^p \right)^{1/p}$$



Wasserstein Stability

Theorem (Stability for Wasserstein distance)

Let $f, g : K \rightarrow \mathbb{R}$ be two simplex-wise monotone functions on a simplicial complex K . Then,

$$d_{W,q}(\text{Dgm}_p(\mathcal{F}_f), \text{Dgm}_p(\mathcal{F}_g)) \leq \|f - g\|_q.$$

What's the difference?

...To the jupyter notebook!!!